

nabta  WELLNESS PARK

THE LIVING SHADE LOOP

DUBAI MUNICIPALITY • AI PARK DESIGN CHALLENGE • DRAFT COMPETITION ENTRY

D•08

Accessibility, Circulation & Commercial Facilities

W08 Accessibility

CSF Map



Deliverable 8 — Accessibility & Circulation Strategy + Indicative Commercial & Service Facilities Plan

A. CIRCULATION HIERARCHY — MOVEMENT FOR EVERY BODY

In a wellness park, circulation *is* the programme: walking, wheeling and jogging are Nabta's primary health offer. One legible spine and two subordinate layers ensure every visitor — stroller, wheelchair user, jogger — can complete the loop with dignity.

- **Primary: the Living Shade Loop.** 1.0 km figure-eight, 3.5 m shared path, continuously shaded. A 1.0 m jogger lane is marked in terracotta-pigmented resin on the outer edge; cross-slope $\leq 2\%$ throughout, gradients $\leq 1:21$, no ramps anywhere.
- **Secondary: garden paths.** 1.8 m stabilised, firm and slip-resistant paths entering each of the ten garden rooms, with 1.8×1.8 m passing/turning bays at every junction.
- **Tertiary: desire-line shortcuts.** Compacted decomposed-granite links across the loop's waist and lawn edges, formalising natural shortcuts; all step-free.

Walk times (1.0 m/s accessible pace): entry plaza → play dune 3 min; → sensory wadi 5 min; full loop 17 min walking, 6 min jogging. No point is more than 4 minutes from a loop exit gate.

B. UNIVERSAL ACCESSIBILITY — CODE-MAPPED CHECKLIST

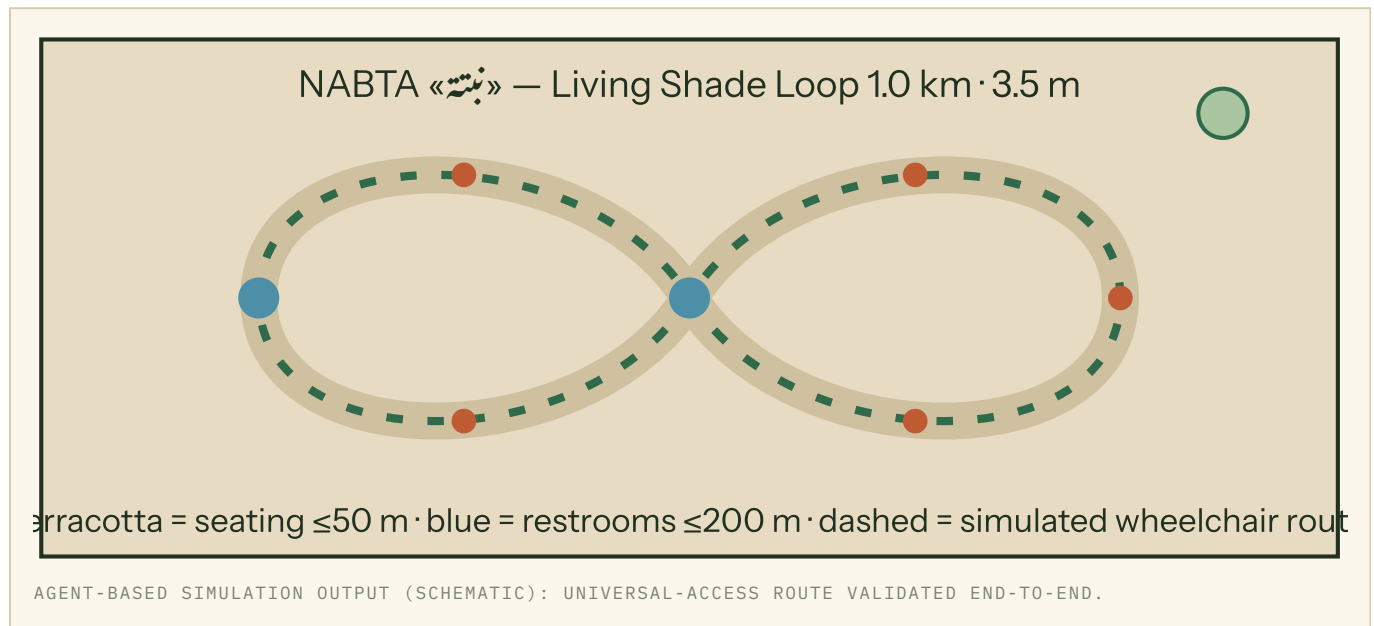
DUBAI UNIVERSAL DESIGN CODE THEME	NABTA MEASURE
Step-free movement	100% step-free park: zero kerbs on the loop, flush terracotta bridges over the wadi ($\leq 1:21$), tactile-edged not stepped.
Wayfinding	Continuous tactile guidance strip on the full 1.0 km loop; tactile/high-contrast decision studs at all ten room junctions.
Rest & comfort	Seating every 50 m maximum, all with armrests and backrests; 50% in year-round shade; adjacent wheelchair pause space at each bench.
Arrival	4 accessible parking bays nearest the SE entry (3.6 m wide, shaded, ≤ 30 m to gate), level drop-off with blister paving.
Sanitary provision	One Changing Places-standard restroom (12 m ² , ceiling hoist, adult changing bench) at the entry plaza block, plus ambulant-accessible cubicles at both restroom locations.
Sensory inclusion	Quiet room directly adjacent to the sensory wadi garden; hearing (induction) loop at the majlis plaza; predictable acoustic zoning.
Signage	Braille + pictogram + high-contrast signage in Arabic and English at all entrances and room thresholds, mounted 1.2–1.6 m.
Inclusive play	Play dune equipment with transfer platforms at 450 mm, wheelchair-accessible carousel, ground-level play at every cluster.

Each row cites its Code theme for line-by-line audit; all dimensions meet or exceed Code minimums.

C. AI-ASSISTED CIRCULATION VALIDATION (HUMAN-LED)

The plan was stress-tested with an **agent-based pedestrian simulation**: ~5,000 synthetic visitors per scenario, with calibrated wheelchair (0.8 m/s), stroller, elderly, child and jogger profiles across Friday-peak, school-run and summer-evening demand. AI *verified*, humans *decided*:

- Confirmed no crossing point where jogger and wheelchair flows conflict at >0.3 encounters/min; the terracotta jogger lane was moved to the outer edge as a result.
- Validated that every simulated wheelchair agent completes the loop with rest seating ≤50 m apart and a restroom within 200 m from any point — counts are simulation-verified, not rules-of-thumb.
- Predicted desire lines matched the tertiary shortcut layer; two AI-suggested shortcuts were **rejected** by the design team on CPTED sightline grounds, logged in the decision record.



D. ARRIVAL STRATEGY

- **Drop-off:** lay-by off the SE access road per RTA geometry — 3 vehicle lengths, level boarding, tactile paving, shade canopy — sized for the school-run peak.
- **Parking:** ~40 car bays (incl. 4 accessible + 2 family) in a shaded grove lot; permeable paving.
- **Public transport:** signed, shaded, step-free links to 2 RTA bus stops within 300 m.
- **Micromobility:** 30 shaded bicycle racks at each of the three entrances (90 total), plus an e-scooter dock per entrance (SE, NW, NE), all outside pedestrian desire lines.

E. SERVICE & EMERGENCY ACCESS

- Loop segments engineered as a **4 m clear emergency route** (path + reinforced shoulder), giving Civil Defence and ambulance reach to all ten rooms; removable bollards at the three gates.
- **W-edge service yard:** screened compound with café back-of-house delivery, maintenance store, and irrigation plant; service vehicles never cross visitor space during opening hours.
- Six paired **waste/recycling points** on the loop, serviced from the W yard before 7:00 am.

F. INDICATIVE COMMERCIAL & SERVICE FACILITIES PLAN

Catchment reasoning: Al Safa 2 is a low-rise villa district — several thousand residents within a 10-minute walk, schools south, Sheikh Zayed Road offices minutes away. Demand peaks at dawn/evening wellness hours,

the school run and winter weekends; fixed commercial space is sized to the reliable base load, flexible cart pads absorb peaks.

FACILITY	LOCATION (GARDEN ROOM)	CATCHMENT LOGIC
Café + shaded terrace (120 m²)	SE entry plaza	Captures school-run parents, loop finishers, and street footfall; visible from drop-off.
2 kiosk cart pads (power/water)	Event lawn edge	Activated during events and weekends; zero fixed cost when idle.
Coffee/juice kiosk	Loop waist, by play dune	Serves caregivers with sightlines to all play zones.
Restrooms ×2 (incl. Changing Places)	Entry plaza + loop waist	No point in the park >200 m from a restroom.
Prayer room + wudu	Beside majlis plaza	Adjacent to gathering space; on tactile route.
Water refill + shade shelters	Every second room	Summer walkability; reduces single-use plastic.

Activation & concession note: café and kiosks are tendered as one concession with revenue share tied to mandated opening hours (06:00–23:00), guaranteeing dawn/evening service through summer when the misted, shaded terrace keeps the SE plaza usable; cart pads flex to winter events without permanent build-out. A wellness clause obliges the operator to a healthy-menu baseline and free water refills, aligning commercial income with the park’s health mission and keeping the commercial footprint under 2% of park area.

G. SAFETY BY DESIGN

- **CPTED:** clear sightlines along the loop, understorey planting below 0.8 m at path edges, no dead-end paths, active café frontage at the main entrance.
- **Adaptive lighting:** loop 20 lux, garden rooms 10 lux, planted gardens 5 lux; all 2700 K warm white, full-cutoff dark-sky-compliant fixtures, presence-dimmed after 23:00.
- **Privacy commitment:** operational CCTV at entrances only; **no facial recognition** and no biometric analytics anywhere in the park.